

7 Osmoreceptors are specialised cells that respond to changes in the water potential of the blood.

0 7 . 1 Give the location of osmoreceptors in the body of a mammal. **[1 mark]**

0 7 . 2 When a person is dehydrated, the cell volume of an osmoreceptor decreases. Explain why. **[2 marks]**

0 7 . 3 Stimulation of osmoreceptors can lead to secretion of the hormone ADH. Describe and explain how the secretion of ADH affects urine produced by the kidneys. **[4 marks]**

[Extra space] _____

The efficiency with which the kidneys filter the blood can be measured by the rate at which they remove a substance called creatinine from the blood. The rate at which they filter the blood is called the glomerular filtration rate (GFR).

In 24 hours, a person excreted 1660 mg of creatinine in his urine. The concentration of creatinine in the blood entering his kidneys was constant at 0.01 mg cm^{-3} .

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 Calculate the GFR in $\text{cm}^3 \text{ minute}^{-1}$.

[1 mark]

Answer = _____

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 Creatinine is a breakdown product of creatine found in muscle tissues. Apart from age and gender, give **two** factors that could affect the concentration of creatinine in the blood.

[1 mark]

1 _____

2 _____

Turn over for the next question

